

SHASHWAT SINGH

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I am currently a PhD student at the University of Glasgow and working on probing massive and supermassive black holes using gravitational wave sources to be detected by LISA – a future space-based gravitational-wave detector and electromagnetic transients from ZTF & LSST.

ACADEMIC QUALIFICATIONS

2023	University of Glasgow, SCHOOL OF PHYSICS AND ASTRONOMY, PhD
2027	Thesis title : Revealing the family of massive black holes with LISA Supervisors : Dr Christopher Berry and Dr John Veitch Gravitational Waves LISA Super/Massive Black Holes Population Inference
2022	l'Observatoire de Paris, UNIVERSITÉ PSL (PARIS SCIENCES & LETTRES), M2 – International Research Track
2023	Program combined with courses and research in laboratories. General-relativity Data-analysis Magneto-hydrodynamics High-performance-computing
2021	Sciences et Ingénierie, SORBONNE UNIVERSITÉ, M1 – Paris Physics Masters
2022	Program targeted towards fundamental courses and compulsory lab-work. Advanced quantum mechanics Statistical mechanics Astrophysics & Cosmology Numerical methods
2017	Sardar Vallabhbhai National Institute of Technology, B. TECH, Mechanical Engineering
2021	Four-year program combined with theoretical and experimental work. Data-analysis Fluids - mechanics & dynamics Machines and rigid body motion

PUBLICATIONS AND PREPRINTS[†]

- › “Gravitational Wave Lensing as a Probe to Understanding Subhaloes Hosting Massive Black Holes”, Kanu, A., Singh, S., Harikumar, S., *In prep.*
- › “Disrupted Stars and Falling Compact Objects : Combining Multi-Messenger Observations to Constrain the Population of Massive Black Holes”, Singh, S., Berry, C. P. L., Broggi, L., Veitch, J., Volonteri, M., *In prep.*
- › “Revealing Massive Black Hole Astrophysics : The Potential of Hierarchical Inference with Extreme Mass-Ratio Inspiral Observations”, Singh, S., Chapman-Bird, C. E. A., Berry, C. P. L., Veitch, J., *The Astrophysical Journal*, 1005, 176 (2026). doi:10.3847/1538-4357/ae75dc
- › “The Early Career Workshop of GR-Amaldi 2025”, Al-Shammari, S. *et al.*, *Journal of Physics : Conference Series*, 3177, 012001 (2026). doi:10.1088/1742-6596/3177/1/012001
- › “Constraints on the Extreme Mass-Ratio Inspiral Population from LISA Data”, Singh, S., Chapman-Bird, C. E. A., Berry, C. P. L., Veitch, J., *Journal of Physics : Conference Series*, 3177, 012113 (2026). doi:10.1088/1742-6596/3177/1/012113
- › “Gravitational Wave Lensing : Probing Fuzzy Dark Matter with LISA”, Singh, S., Brando, G., Savastano, S., Zumalacárregui, M., *Journal of Cosmology and Astroparticle Physics*, 2025, 025 (July 2025). doi:10.1088/1475-7516/2025/07/025
- › “Estimating Dynamical Parameters of Two Interacting Galaxies Using Deep Learning”, Mahor, A., Reddy, J., Singh, A., Singh, S., *Monthly Notices of the Royal Astronomical Society*, 521(3), 3441–3450 (May 2023). doi:10.1093/mnras/stad700
- › “Deep Learning for Estimating Parameters of Gravitational Waves”, Singh, S., Singh, A., Prajapati, A., Pathak, K. N., *Monthly Notices of the Royal Astronomical Society*, 508(1), 1358–1370 (November 2021). doi:10.1093/mnras/stab2417
- › “Lindblad Evolution and Quantum to Classical Transition of Rabi Oscillation in Single Quantum Dot”, Prajapati, A., Singh, S., *AIP Conference Proceedings*, 2220, 020122 (2020). doi:10.1063/5.0001258
- › “Experimental and Numerical Investigation of Thermal Performance of Synthetic Jet Impingement”, Singh, P. K., Kothar, R., Sahu, S., Upadhyay, P. K., Singh, S., *Proceedings of the ASME 2020 28th International Conference on Nuclear Engineering (ICONE28)*, V001T03A020 (2020), 6 pages. doi:10.1115/ICONE2020-16775
- › “Experimental and Numerical Investigation of the Thermal Performance of Impinging Synthetic Jets with Different Waveforms”, Singh, P. K., Sahu, S., Upadhyay, P. K., Singh, S., *Experimental Heat Transfer* (2021). doi:10.1080/08916152.2021.1984341
- › “Flow and Heat Transfer Analysis of an Axisymmetric Impinging Synthetic Jet for Electronic Cooling”, Singh, P. K., Kumar, A., Shah, A., Kishor, A., Sahu, S. K., Upadhyay, P. K., Singh, S., *Proceedings of the International Conference on Innovation and Thermo-Fluid Engineering and Science (ICITFES-2020)*, NIT Rourkela, India, 10–12 February 2020. [Paper ID : 13754]

- “Decoherence Control via Pumping of Electromagnetic Energy in Open Quantum System”, Prajapati, A., Singh, S., presented at *The 5th International Conference on Atomic, Molecular, Nano-physics with Application (CAMNP-2019)*.
- “Predicting Future Astronomical Events Using Deep Learning”, Singh, S., Prajapati, A., Pathak, K. N. arXiv:2012.15476.[†]
- “Prospects of Detection of Lensed Gravitational Wave Signals”, Singh, S. (2022). *Zenodo*. doi:10.5281/zenodo.7029226.[†]

CONFERENCES, SEMINARS & WORKSHOPS[‡] CONTRIBUTIONS

- National Astronomy Meeting 2026, University of Birmingham, July 2026. *Inferring MBH's origin and growth with multimessenger observation from EMRI and TDE observations*.
- European Astronomical Society Annual Meeting, SwissTech Convention Centre Lausanne, Switzerland, July 2026. *Probing massive black hole population with combined EMRI and TDE observations*.
- Massive Black Hole Spin Workshop, University of Edinburgh, UK, April 2026. *Disrupted stars and falling compact objects : Combining multi-messenger observations to constrain the population of MBH*.
- GW:UK@Nottingham, University of Nottingham, January 2026. *The potential of hierarchical inference with extreme mass-ratio inspiral observations*.
- Reunion populations de sources GDR ondes gravitationnelles, Institut d'Astrophysique de Paris, September 2025. *Revealing the massive black hole population with LISA*.
- GR24 & Amaldi16, University of Glasgow, UK, July 2025. *Constraints on EMRI Populations from LISA data*.
- EMRI Search & Inference within the LISA Global Fit - Part I, APC Paris, France, June 2025. *Accelerated MBH population estimation from EMRI detections*.
- SUPA Cormack, University of Dundee, UK December 2024. *Constraining Massive Black Hole population from LISA observations*.
- LISA Astrophysics Working Group Meeting, MPA Garching, Germany, Nov 2024. *Constraining EMRI Population using Hierarchical Bayesian Inference*.
- Cosmology from Home, July 2023. *Probing Fuzzy dark matter with lensed Gravitational waves*.
- 3rd International Conference on Condensed Matter and Applied Physics (ICC) - Bikaner, India, October 2019. *Lindblad Evolution and Quantum to Classical Transition of Rabi Oscillation in Single Quantum Dot*.
- 9th International Conference on Gravitation and Cosmology (ICGC) - IISER Mohali, India, December 2019. *Clustering and Predicting Astrophysical events using GW*.
- Python for HPC Workshop by Max Planck Computing and Data Facility (MPCDF)[‡], July 2023
- 2nd MaNiTou Summer School on Gravitational Waves : A new window to the Universe[‡] - Nice, France, July 2023.

CONFERENCES, SEMINARS & WORKSHOPS ORGANISATION

- GR24 & Amaldi16, University of Glasgow, UK, July 2025
- Student Seminar, University of Glasgow, UK, 2024–2025

INVITED TALKS

- Physik-Institut, Universität Zürich, January 2026. *Revealing MBH astrophysics using EMRIs*.
- Institute for Gravitational Research, January 2026. *Revealing the population of massive black holes*.
- LISA Working Group, January 2026. *Constraints on EMRI Populations from LISA data*.
- PyCBC Telecon, April 2025. *Gravitational wave lensing : probing Fuzzy Dark Matter with LISA*.
- LISA Cosmology Working Group, April 2025. *Gravitational wave lensing : probing Fuzzy Dark Matter with LISA*.

AWARDS AND RECOGNITION

2023-26	University of Glasgow Graduate School Mobility Grant
2025	Lord Kelvin or Charles Lindie Mitchell Scholar
2023	University of Glasgow Graduate School scholarship.
2023	Received Erasmus+ funding (EU's program to support education, training, youth and sport in Europe)
2022	Université PSL fellowship for higher education (M2).
2022	Received IPIASMUS grant (IPI Initiative « Physique des Infinis ») towards carrying out an internship at Max-Planck-Institut für Gravitationsphysik (Albert-Einstein-Institut).
2020	Received honorarium by the Max-Planck-Institut für Gravitationsphysik (Albert-Einstein-Institut) for a three-month internship.

RESEARCH EXPERIENCE

- Master Thesis, Max-Planck-Institut für Gravitationsphysik (*Albert-Einstein-Institut*), (Duration : January 2023 – July 2023; Supervisor : Dr M. Zumalacárregui) – Probing Fuzzy dark matter using lensed gravitational waves detected by LISA.
- Lab Insertion Unit, l'Observatoire de Paris, (Duration : September 2022 – January 2023; Supervisor : Dr A. Hees (SYRTE) & Dr N. Korsakova (APC)) – Waveform compression of Extreme-Mass-Ratio-Inspirals (EMRIs) using Singular Value Decomposition.

- Internship, Max-Planck-Institut für Gravitationsphysik (*Albert-Einstein-Institut*), (Duration : April 2022 – July 2022; Supervisor : Dr A. H. Nitz) – Prospects of premerger detections & sky-localization of gravitational waves (GWs), using higher modes.
- Internship, Max-Planck-Institut für Gravitationsphysik (*Albert-Einstein-Institut*), (Duration : June 2020 – September 2020; Supervisor : Dr A. H. Nitz) – Built a prototype analysis for massive binary black hole (MBH) mergers using LISA.
- Dept. of Physics, SVNIT Surat, (Duration : May 2019 – August 2020; Under supervision of Dr K. N. Pathak) – Deep learning for estimating parameters of GWs (CNNs) and sequence-prediction of galaxy mergers (LSTM networks).
- Internship, Dept. of Mechanical Engineering, Indian Institute of Technology Indore, (Duration : May 2019 – July 2019; Supervisor : Dr S. K. Sahu) – Developed a numerical method to study heat transfer effects of a synthetic jet on different materials, and C++ code allowing mesh motion within the model in Ansys Fluent.
- Visiting researcher, Max-Planck-Institut für Gravitationsphysik (*Albert-Einstein-Institut*), (Duration : January 01 – January 09, 2018; Supervisor : Dr M. Zumalacárregui)
- Visiting researcher, Institut Astrophysique de Paris, (Duration : January 01 – January 09, 2018; Supervisor : Prof Marta Volonteri)

</> PROGRAMMING & TECHNICAL SKILLS

LANGUAGES & TOOLS : Python, C++/C, Mathematica, SQL

SCIENTIFIC COMPUTING : ML / DL, HPC

CLOUD TECHNOLOGIES : AWS S3, CDN, λ

FRONT-END & BACK-END : React + JS, HTML, CSS

📖 PUBLIC LIBRARIES

MBHPOPULATION

2024-PRESENT

🔗 github.com/SSingh087/MBHpopulation

Unveiling the population of massive black holes

gravitational-waves electromagnetic waves massive black hole LISA ZTF population inference

POPLAR

2024-2025

🔗 github.com/cchapmanbird/poplar

Augmenting selection bias modelling with machine learning for big speedups

gravitational-waves massive black hole LISA population inference deep learning

GLOW

2023

🔗 github.com/miguelzuma/GLOW_public

Contributed to the GW lensing library

gravitational-waves lensed-gravitational-waves

INCLUSION OF HIGHER MODES FOR PARAMETER ESTIMATION OF GRAVITATIONAL WAVES IN PyCBC

2022

🔗 github.com/SSingh087/pycbc/tree/conmodel

A recovery model that allows extracting mode-by-mode information while performing parameter estimation.

gravitational-waves higher-modes premerger-detection multi-messenger astronomy PyCBC

LENSGW FOR GENERATING GRAVITATIONALLY LENSED SIGNALS

2021

🔗 github.com/SSingh087/lensGW

Python library for generating lensed gravitational waves and uses PyCBC for waveform generation so that all analysis can be done using tools provided by PyCBC.

gravitational-waves lensed-gravitational-waves PyCBC

LENSGW-PyCBC-PLUGIN

2021

🔗 github.com/SSingh087/lensGW-PyCBC-plugin

Plugin for allowing waveforms to be recognised by PyCBC and perform parameter estimation

gravitational-waves lensed-gravitational-waves PyCBC

LISA - MODULE

2020

🔗 github.com/gwastro/pycbc/commits/master/pycbc/detector.py?author=SSingh087

Prototype for analysis of MBH GWs signals using LISA space-based GW observatory.

The module consists of a simplified LISA orbit and detector response towards a GW signal.

gravitational-waves LISA PyCBC

- 2024, Physics Lab, P2.

Undergraduate[○] and master's* students (via Bose.X)

- 2024, Nancy Sharma[○] & Sree Suswara*, SVNIT, Surat, *Gravitational-Wave Lensing + Machine Learning*.
- 2023, Nancy Sharma[○] Nancy Jikarda[○], SVNIT, Surat, *Gravitational-Wave Lensing*.
- 2021, Adarsh Mahor[○] & Janvita Reddy[○], SVNIT, Surat, *Co-authored a paper; Application of ML in astronomy*.
- 2025, Ankita Chaudhary*, Osmania University,
- 2025, Ansal Kanu[○], University of Delhi, *Co-authored a paper; GW lensing*.
- 2025, Ananthu K Lali*, University of Alabama, Huntsville,
- 2025, Shravan*

EXTRACURRICULAR & OUTREACH ACTIVITIES

- BOSE.X, Co-FOUNDER : Independent research organization for undergraduate research; since 2019. (bosex.org).
- GRA-WITTY, FOUNDER : Outreach program; since 2023. (gra-witty)
- GLASGOW SCIENCE FESTIVAL : Outreach activity, 2024
- WORLDCON : Outreach activity, 2024